

Problem 1. Show that the additive inverse, or negative, of an even number is an even number using a direct proof.

Answer.

An even integer n can be represented by $n = 2k$, where k is any integer.

$$\begin{aligned} -n &= -2k \\ -n &= 2(-k) \end{aligned}$$

Since k is an integer, $-k$ is also an integer. let's call this integer m .

$$\begin{aligned} -n &= 2(-k) \\ -n &= 2m \quad (\text{substitution}) \end{aligned}$$

Since $-n$ can be written in the form $2m$, and both m and k is an integer, $-n$ is an even number by definition. ■

Problem 2. Prove that if n is a perfect square, then $n + 2$ is not a perfect square.

Answer.

A perfect square is defined as a number k square. so, if n is a perfect square, n can be represented as $n = k^2$, where k is a non-negative integer.

If $n + 2$ is also a positive square, it should be after n . We can find the distance between the original number for it to be a perfect square:

$$\begin{aligned}(k + 1)^2 - k^2 \\&= k^2 + 2k + 1 - k^2 \\&= 2k + 1\end{aligned}$$

This problem states that $n + 2$ is not a perfect square.

$$\begin{aligned}2k + 1 &= 2 \\2k &= 1 \\k &= 0.5\end{aligned}$$

However, k have to be an integer for n and $n + 2$ to be perfect squares. This supports the statement in the problem. ■

Problem 3. Prove that if x , y , and z are integers and $x + y + z$ is odd, then at least one of x , y , and z is odd.

Answer.

By the contrapositive law, we negate the implication. We get: “If x , y , and z are integers and $x + y + z$ is *even*, then *for all of* the possibilities of x , y , and z , they are even.”

We can represent an even number $n = 2k$ for an arbitrary integer k . Therefore, we can let $x = 2k$, $y = 2j$, $z = 2h$:

$$\begin{aligned}x + y + z & \\ &= 2k + 2j + 2h \quad (\text{substitution}) \\ &= 2(k + j + h)\end{aligned}$$

since k , j , and h are all integers, we call the sum of these integers m . ($m = k + j + h$)

$$\begin{aligned}2(k + j + h) & \quad (\text{from previous}) \\ &= 2m \quad (\text{substitution})\end{aligned}$$

since $m = k + j + h$ is an integer, the sum of $x + y + z = 2m$ is even by definition. therefore, by contrapositive law, we know that for the integers x , y , and z , at least one variable must be odd. ■

Problem 4. Prove that $m^2 = n^2$ if and only if $m = n$ or $m = -n$.

Answer.

Proposition: $\forall m \forall n ((m^2 = n^2) \Leftrightarrow ((m = n) \vee (m = -n)))$

Since this is a bidirectional proposition, we have to prove two parts.

First implication: $\forall m \forall n (m^2 = n^2) \rightarrow ((m = n) \vee (m = -n))$

$$\begin{aligned} m^2 &= n^2 \\ m^2 - n^2 &= 0 \\ (m - n)(m + n) &= 0 \end{aligned}$$

If a product of two variables equals to 0, that means at least one of the multiplier have to be 0. If $(m - n) = 0$, that means $m = n$; and if $(m + n) = 0$, that means $m = -n$.

Second implication: $\forall m \forall n ((m = n) \vee (m = -n)) \rightarrow (m^2 = n^2)$

First *or* case: $\forall m \forall n (m = n) \rightarrow (m^2 = n^2)$

$$\begin{aligned} m &= n \\ m^2 &= n^2 \quad (\text{square both sides}) \end{aligned}$$

Second *or* case:

$$\begin{aligned} m &= -n \\ m^2 &= (-n)^2 \quad (\text{square both sides}) \\ m^2 &= n^2 \quad (\text{square will always be positive}) \end{aligned}$$

Since we've proven the bidirectional proposition is true in both ways, the statement is proven to be true. ■

Problem 5. Show that the product of two of the numbers $65^{1000} - 8^{2001} + 3^{177}$, $79^{1212} - 9^{2399} + 2^{2001}$, and $24^{4493} - 5^{8192} + 7^{1777}$ is non-negative. Is your proof constructive or non-constructive? [Hint: Do not try to evaluate these numbers!]

Answer. Since the result is non-negative, that means the two multipliers must have the same sign. This is because if the two multipliers (the huge sum of the expression) have mixed signs (including zero), the result would be negative.

Let the two numbers (result of the addition) be M and N .

Case 1: $M \geq 0$ and $N \geq 0$. The product of two non-negative numbers are non-negative.

Case 2: $M < 0$ and $N < 0$. The product of two negative numbers are positive (which is non-negative).

If either M or N is zero, the product would be zero, also non-negative. ■

Problem 6. Show that if r is an irrational number, there is a unique integer n such that the distance between r and n is less than $\frac{1}{2}$.

Answer.

We find the contrapositive: “If there is no integer n such that the distance between r and n is less than $\frac{1}{2}$, or if there are at least two such integers, then r is a rational number.”

Since we have an *or* statement, we consider two cases.

Case 1: we assume there is no integer n where $|r - n| < \frac{1}{2}$. The distance between two integers must be at least one, and there are no possibilities where an integer would have a distance where it is less than $\frac{1}{2}$ because it must be either on the left, right, (two integers, 2nd case) or directly in the middle (unique number resulting in 0).

Case 2: there are two or more integers within a distance of less than $1/2$.

Problem 7. The **quadratic mean** of two real numbers x and y equals $\sqrt{\frac{x^2+y^2}{2}}$. By computing the arithmetic and quadratic means of different pairs of positive real numbers, formulate a conjecture about their relative sizes and prove your conjecture.

Answer.

Proposition: for any real number x and y , the quadratic mean $\sqrt{\frac{x^2+y^2}{2}}$ will always be positive.

Proof: Because both x and y are real numbers, their square should be non-negative, since the square of all real numbers are positive, and the square of zero is zero. Therefore, $x^2 + y^2$ is also non-negative. The division of a non-negative number by two is either a positive number, or zero. Therefore, the result of the division is non-negative is well. As for the final operation (square root), the square root of zero is zero, and the square root of a positive number will always be a positive number. ■

Problem 8. Prove that there are no solutions in integers x and y to the equation $2x^2 + 5y^2 = 14$.

Answer.

Challenge 1. Prove that $\sqrt[3]{2}$ is irrational.

Answer. We assume $\sqrt[3]{2}$ is rational. Therefore, it can be represented as

$$\sqrt[3]{2} = \frac{a}{b}$$

Where a and b are integers with no common factors other than 1, and $b \neq 0$.

$$\begin{aligned}\sqrt[3]{2} &= \frac{a}{b} \\ 2 &= \frac{a^3}{b^3} \\ 2b^3 &= a^3\end{aligned}$$

Since b is an integer, b^3 will be an integer as well. By definition, $2b^3$ where b^3 is an integer results in an even number. This shows that a^3 is an even integer as well, for an arbitrary integer k , where $a^3 = 2k$.

$$\begin{aligned}2b^3 &= a^3 \\ 2b^3 &= (2k)^3 && \text{(substitution)} \\ 2b^3 &= 8k^3 \\ b^3 &= 4k^3 = 2(2k^3)\end{aligned}$$

We can get that b^3 is an even integer as well, since $2k^3$ is an arbitrary integer. We now know that b^3 and a^3 are both even, which by definition, will have a common factor of 2. However, in the definition that we wrote for $\frac{a^3}{b^3}$ above, the fraction should be in its lowest terms, where a^3 and b^3 have no common factors other than 1. This contradicts our original assumption where $\sqrt[3]{2}$ is rational, showing that it is irrational. ■